Cyber Security Task 2 (2026)

Phishing Email Detection & Awareness System

1. Introduction

Phishing is one of the most common cyber-attacks used by attackers to steal sensitive information such as login credentials, financial details, and personal data. In phishing attacks, cybercriminals impersonate trusted organizations and send fraudulent emails to trick users into clicking malicious links, downloading infected attachments, or sharing confidential information.

Many organizations face significant financial and reputational losses due to phishing attacks. Most successful phishing attacks occur not because of weak security systems, but because users fail to recognize suspicious emails.

This project focuses on analysing phishing email samples, identifying phishing indicators, classifying the risk level of emails, and creating awareness guidelines that help users identify and avoid phishing attacks.

---

2. Objective of the Project

The main objectives of this project are:

- To analyze phishing email samples.
- To identify common phishing indicators in emails.
- To examine sender domains and suspicious URLs.
- To classify the risk level of phishing emails.
- To create awareness guidelines for employees and users.
- To educate users on how to detect and avoid phishing attacks.

---

3. Tools Used

The following tools were used to analyze phishing emails and investigate suspicious email components.

Tool	Purpose
Google Message Header Analyzer	To analyze email headers and identify sender information
MX Toolbox Email Header Analyzer	To detect suspicious email routing and authentication issues
Browser Inspection Tools	To safely examine URLs and domain links
WHOIS Lookup	To check domain age and domain ownership

These tools help security analysts determine whether an email is legitimate or malicious.

MX TOOLBOX SUPERTOOL

Pricing | Tools | Delivery Center | Monitoring | Products | Blog | Support | Login

SuperTool | MX Lookup | Blacklists | DMARC | Diagnostics | Email Health | DNS Lookup | Analyze Headers | All Tools

Header Analyzed

Email Subject: Urgent: Your Account Will Be Locked

Analyze New Header

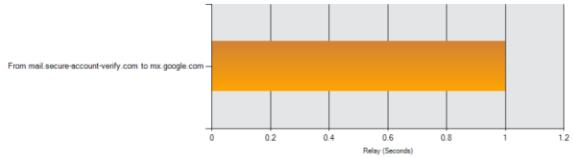
Copy/Paste Warning
Copy/Pasting a header works for most people, but sometimes it can cause problems with things like DKIM Validation. For the best results, use our Email Deliverability tool

Delivery Information

- DMARC Compliant
  - SPF Alignment
  - SPF Authenticated
  - DKIM Alignment
  - DKIM Authenticated

Relay Information

Received Delay:	0 seconds



Hop	Delay	From	By	With	Time (UTC)	Blacklist
1	"	mail.secure-account-verify.com 185.203.119.45	mx.google.com	ESMTP	3/6/2026 5:23:45 PM	✓

Gmail & Yahoo are now requiring DMARC - Get yours setup with Delivery Center

SPF and DKIM Information

dmarc:secure-account-verify.com Show | Solve Email Delivery Problems

```
v=DMARC1; p=quarantine; adkim=s; aspf=s;
```

spf:secure-account-verify.com:185.203.119.45 Hide

```
v=spf1 mx -all
```

Prefix	Type	Value	PrefixDesc	Description
	v	spf1		The SPF record version
+	mx		Pass	Match if IP is one of the MX hosts for given domain name.
-	all		Fail	Always matches. It goes at the end of your record.

	Test	Result	
✗	SPF Authentication	SPF Failed for IP - 185.203.119.45	More Info
✓	SPF Record Published	SPF Record found	
✓	SPF Record Deprecated	No deprecated records found	
✓	SPF Multiple Records	Less than two records found	
✓	SPF Alignment	Domain found in SPF	
✓	SPF Contains characters after ALL	No items after 'ALL'.	
✓	SPF Syntax Check	The record is valid	
✓	SPF Included Lookups	Number of included lookups is OK	
✓	SPF Recursive Loop	Nor Recursive Loops on Includes	
✓	SPF Duplicate Include	No Duplicate Includes Found	
✓	SPF Type PTR Check	No type PTR found	

Dkim Signature Error:
No DKIM-Signature header found - more info

Dkim Signature Error:
There must be at least one aligned DKIM-Signature for the message to be considered aligned. - more info

Headers Found

Header Name	Header Value
Delivered-To	user@example.com
Received-SPF	fail (google.com: domain of security@secure-account-verify.com does not designate 185.203.119.45 as permitted sender)
Authentication-Results	mx.google.com; spf=fail smtp.mailfrom=security@secure-account-verify.com
From	Security Team <security@secure-account-verify.com>
To	user@example.com
Subject	Urgent: Your Account Will Be Locked
Date	Fri, 06 Mar 2026 10:23:40 -0700
Message-ID	<123456789@mail.secure-account-verify.com>
MIME-Version	1.0
Content-Type	text/html

Received Header

```
Delivered-To: user@example.com
Received: from mail.secure-account-verify.com (mail.secure-account-verify.com [185.203.119.45])
        by mx.google.com with ESMTP id x12si1234567qka.123.2026.03.06.10.23.45
        for <user@example.com>;
        Fri, 06 Mar 2026 10:23:45 -0700 (PDT)
Received-SPF: fail (google.com: domain of security@secure-account-verify.com does not designate 185.203.119.45 as permitted sender)
Authentication-Results: mx.google.com;
       spf=fail smtp.mailfrom=security@secure-account-verify.com
From: Security Team <security@secure-account-verify.com>
To: user@example.com
Subject: Urgent: Your Account Will Be Locked
Date: Fri, 06 Mar 2026 10:23:40 -0700
Message-ID: <123456789@mail.secure-account-verify.com>
MIME-Version: 1.0
Content-Type: text/html
```

Permanently forget this email header

---

4. Phishing Email Sample

Below is a sample phishing email used for analysis.

Subject: Urgent: Your Account Will Be Locked

Email Body

Dear User,

We noticed suspicious activity on your account. To avoid account suspension, please verify your account details immediately.

Verify Now:
http://secure-account-verify[.]com

Failure to verify within 24 hours will result in permanent account lock.

Regards
Security Team

---

5. Email Header Analysis

Email headers provide technical information about how the email was transmitted, including sender IP address, authentication results, and server routing.

The header was analyzed using **Google Header Analyzer and MXToolbox**.

Observations

Header Field	Observation
Sender Email	security@secure-account-verify.com
Sender Domain	Unknown domain

Header Field	Observation
SPF Authentication	Failed
Sender IP Address	Suspicious external IP
Domain Reputation	Newly registered domain

Analysis Result

The header analysis shows that the sender domain is not trusted and authentication checks failed. This indicates that the email may have been sent from a fraudulent or spoofed source.

---

6. Phishing Indicators Identified

Several phishing indicators were identified in the email.

Indicator	Description
Generic Greeting	Uses “Dear User” instead of the recipient’s name
Urgent Language	Creates panic by threatening account suspension
Suspicious Link	URL does not belong to any legitimate organization

Indicator	Description
Fake Sender	Sender domain appears suspicious
Fear Tactics	Forces users to act quickly

These indicators strongly suggest that the email is a phishing attempt.

---

7. URL and Domain Analysis

The phishing email contains the following link:

http://secure-account-verify[.]com

Issues Identified

- The domain name appears suspicious and does not belong to a legitimate organization.
- The website uses **HTTP instead of HTTPS**, indicating a lack of security.
- The domain name is designed to appear trustworthy by including words such as *secure* and *verify*.

Attackers commonly create fake domains that resemble legitimate websites to trick users.

8. Risk Classification

Based on the email analysis, the risk level was classified using the following parameters.

Parameter	Result
Sender Authenticity	Suspicious
Link Safety	Malicious
Language Pattern	Phishing style
Domain Reputation	Untrusted

Final Risk Classification

High Risk – Phishing Email

Users interacting with this email may expose sensitive information such as login credentials or financial details.

---

9. How the Phishing Attack Works

In this phishing attack, the attacker impersonates a security team and sends an urgent email claiming suspicious activity on the user's account.

The email pressures the user to click a verification link to prevent account suspension. Once the user clicks the

link, they may be redirected to a fake login page designed to capture their credentials.

After obtaining the user's credentials, attackers can access accounts, steal sensitive information, or perform financial fraud.

---

10. Phishing Prevention Guidelines

To prevent phishing attacks, users should follow these security practices.

Verify the Sender

Always check the sender's email address and domain before trusting the message.

Avoid Clicking Suspicious Links

Hover over links to verify the actual destination before clicking.

Be Careful with Urgent Messages

Emails that create urgency or panic should be treated with suspicion.

Do Not Share Sensitive Information

Legitimate organizations never request passwords or confidential information via email.

Report Suspicious Emails

If an email appears suspicious, report it immediately to the organization's IT or security team.

---

11. Employee Awareness: Do’s and Don’ts

Do’s

- Verify the sender’s email address.
- Check links before clicking.
- Report suspicious emails.
- Enable multi-factor authentication.

Don’ts

- Do not click unknown links.
- Do not download attachments from unknown senders.
- Do not share passwords or OTPs via email.
- Do not respond to suspicious emails.

---

12. Conclusion

Phishing attacks remain one of the most common cybersecurity threats targeting individuals and organizations. These attacks exploit human behavior by creating urgency and impersonating trusted entities.

Through proper awareness, email analysis, and security practices, users can significantly reduce the risk of phishing attacks. Educating employees about phishing indicators and safe email practices plays a critical role in improving organizational cybersecurity.

---

13. References

The following resources were used for educational purposes and research on phishing email analysis.

- https://github.com/rf-peixoto/phishing_pot
- https://github.com/autinerd/phishing-mail-examples
- https://github.com/sadat1971/Phishing_Email
- https://github.com/Phishing-Database/Phishing.Database